BioOne COMPLETE

Experimental Assessment of Lithium Hydride's Space Radiation Shielding Performance and Monte Carlo Benchmarking

Authors: Schuy, Christoph, Tessa, Chiara La, Horst, Felix, Rovituso, Marta, Durante, Marco, et al.

Source: Radiation Research, 191(2) : 154-161

Published By: Radiation Research Society

URL: https://doi.org/10.1667/RR15123.1

BioOne Complete (complete.BioOne.org) is a full-text database of 200 subscribed and open-access titles in the biological, ecological, and environmental sciences published by nonprofit societies, associations, museums, institutions, and presses.

Your use of this PDF, the BioOne Complete website, and all posted and associated content indicates your acceptance of BioOne's Terms of Use, available at www.bioone.org/terms-of-use.

Usage of BioOne Complete content is strictly limited to personal, educational, and non - commercial use. Commercial inquiries or rights and permissions requests should be directed to the individual publisher as copyright holder.

BioOne sees sustainable scholarly publishing as an inherently collaborative enterprise connecting authors, nonprofit publishers, academic institutions, research libraries, and research funders in the common goal of maximizing access to critical research.

Downloaded From: https://bioone.org/journals/Radiation-Research on 02 Nov 2024
Terms of Use: https://bioone.org/terms-of-use Access provided by University College London

RADIATION RESEARCH **191**, 154–161 (2019)
0033-7587/19 $15.00
©2019 by Radiation Research Society.
All rights of reproduction in any form reserved.
DOI: 10.1667/RR15123.1

Experimental Assessment of Lithium Hydride's Space Radiation Shielding Performance and Monte Carlo Benchmarking

Christoph Schuy,[a,1] Chiara La Tessa,[b,c] Felix Horst,[a,d] Marta Rovituso,[b] Marco Durante,[b,c] Martina Giraudo,[e,f] Luca Bocchini,[e,g] Marcello Baricco,[h] Alberto Castellero,[h] Gianluca Fiore[h] and Uli Weber[a]

[a] *GSI Helmholtz Centre for Heavy Ion Research, Darmstadt, Germany;* [b] *Trento Institute for Fundamental Physics and Applications (TIFPA), Povo, Italy;* [c] *Department of Physics, University of Trento, Povo, Italy;* [d] *Institute of Medical Physics and Radiation Protection (IMPS), THM University of Applied Sciences Giessen, Giessen, Germany;* [e] *Thales Alenia Space, Turin, Italy;* [f] *Department of Applied Science and Technology, Politecnico di Torino, Turin, Italy; and Departments of* [g] *Physics and* [h] *Chemistry and NIS, University of Torino, Turin, Italy*

Schuy, C., La Tessa, C., Horst, F., Rovituso, M., Durante, M., Giraudo, M., Bocchini, L., Baricco, M., Castellero, A., Fiore, G. and Weber, U. Experimental Assessment of Lithium Hydride's Space Radiation Shielding Performance and Monte Carlo Benchmarking. *Radiat. Res.* 191, 154–161 (2019).

The harmful effects of space radiation pose a serious health risk to astronauts participating in future long-term missions. Such radiation effects must be considered in the design phase of space vessels as well as in mission planning. Crew radioprotection during long periods in deep space (e.g., transit to Mars) represents a major challenge, especially because of the strong restrictions on the passive shielding load allowed on-board the vessel. Novel materials with better shielding performance compared to the "gold standard" high-density polyethylene are therefore greatly needed. Because of the high hydrogen content of hydrides, lithium hydride has been selected as a starting point for further studies of promising candidates to be used as passive shielding materials. In the current experimental campaign, the shielding performance of lithium hydride was assessed by measuring normalized dose, primary beam attenuation and neutron ambient dose equivalent using 430 MeV/u ^{12}C, 600 MeV/u ^{12}C and 228 MeV proton beams. The experimental data were then compared to predictions from the Monte Carlo transport codes PHITS and GRAS. The experimental results show an increased shielding effectiveness of lithium hydride compared to reference materials like polyethylene. For instance, the attenuation length for 600 MeV/u ^{12}C primary particles in lithium hydride is approximately 20% shorter compared to polyethylene. Furthermore, the comparison results between both transport codes indicates that the standard Tripathi-based total reaction cross-section model of PHITS cannot accurately reproduce the presented experimental data, whereas GRAS shows reasonable agreement. **© 2019 by Radiation Research Society**

INTRODUCTION

When high-energy heavy ions originating from galactic cosmic rays (GCR) (*1*) and solar particle events (SPE) (*2*) interact with a spacecraft, this results in a complex radiation field inside the vessel. Charged and uncharged projectile- and target-like fragments, as well as slowed-down and scattered primary particles, pose a serious risk for an astronaut's health and might ultimately limit manned space explorations (*3*). Therefore, a deep understanding of the complex radiation field originating from the interaction of external radiation with the vessel hull is mandatory for mission planning, risk management and for finding and validating new mitigation strategies. One of the key countermeasures for radiation protection on Earth is the use of passive shielding materials. However, applying the same concept to a space environment is difficult, due to severe weight constraints at launch as well as the broad energy range and species variety of charged particles in space (*4*). The basic idea of passive shielding for energetic charged ions is to either fully stop them via energy loss by ionization or to break them into lower linear energy transfer (LET) particles via nuclear fragmentation to decrease their effective dose (*5*). Achieving a full stop of all the GCR components with only passive shielding is not feasible due to their high penetration power. Therefore, light materials with high hydrogen content are considered the best shielding due to their superior ability to cause nuclear fragmentation of the incoming radiation per unit mass (*6*). Currently, particle transport codes such as the Particle and Heavy Ion Transport System (PHITS) (*7, 8*) and Geant4 (*9–11*) are widely used to study new radioprotection strategies, but their reliability in simulating complex structures is essentially linked to the quality of their physical models and measured input parameters such as nuclear cross sections. Therefore, accelerator-based tests are highly useful for benchmarking the validity and quality of these codes (*12*).

Through the ROSSINI2 project (RadiatiOn Shielding by ISRU and/or Innovative materials for EVA, vehicles and habitats), the goal is to assess the shielding performance of

[1] Address for correspondence: GSI Helmholtz Centre for Heavy Ion Research, Planckstraße 1, 64291 Darmstadt, Germany; email: c.schuy@gsi.de.

Downloaded From: https://bioone.org/journals/Radiation-Research on 02 Nov 2024
Terms of Use: https://bioone.org/terms-of-use Access provided by University College London

TABLE 1
Description of Beam Configurations and Shielding Targets

Experimental setup	Beam	Intensity	Targets
Normalized dose (HIT)	430 MeV/u ^{12}C	10^8 pps	LiH (5), Al (7)
Attenuation of primary beam (GSI)	600 MeV/u ^{12}C	1,500 pps	LiH (6), PE (3)
Neutron ambient dose equivalent (PTC)	228 MeV protons	1.5×10^{10} pps	LiH (3), PE (3), Al (3)

Note. Beam intensity is indicated in particles/s and the number of different target thicknesses used in each experiment is indicated in parentheses in the last column.

innovative single materials or multi-material structures as well as to provide benchmarking data for commonly used particle transport codes. Datasets for various beams, energies and materials were measured, including dose reduction, multiple scattering, neutron production, attenuation of the primary beam flux as well as nuclear fragmentation. Due to its low atomic number, high hydrogen content and the relatively high absorption cross section for thermal neutrons of ^{6}Li, lithium hydride (LiH) was selected as a promising candidate for radiation shielding purposes in space (*13*) and was recently included in several *in silico* studies (*14–16*). One part of the ROSSINI project was an experimental study on dose reduction of different innovative materials and multilayer configurations (*17*), including LiH when exposed to high-energy light and heavy ions. The results indicated that LiH provided a better dose reduction than standard shielding materials, such as polyethylene (PE), and thus more experiments were performed to further characterize its shielding capabilities and to compare them to Monte Carlo predictions.

The data presented here were collected during several experimental campaigns in 2015 and 2016 at the Helmholtz Centre for Heavy Ion Research (GSI; Darmstadt, Germany), the Proton Therapy Center Trento (PTC; Povo, Italy) and the Heidelberg Ion-Beam Therapy Center (HIT; Heidelberg, Germany). Different beam and target configurations were used for measuring normalized dose, attenuation of the primary beam flux and neutron ambient dose in LiH targets as well as reference materials, namely aluminum (Al) and polyethylene (*18*). A summary of the performed measurements is shown in Table 1.

Engineering or safety related issues of LiH, such as different manufacturing processes (*19*), swelling or outgassing of irradiated LiH (*20*), as well as possible dual uses (*21*) or the effects of different compositions of LiH, are not within the scope of this article and thus will not be discussed.

MATERIALS AND METHODS

The experimental setups used during the measurement campaigns are described in this section, followed by a brief summary of the used target geometries. The accelerator settings and targets of each test can be found in Table 1 while the schematics of the experimental setups are shown in Fig. 1.

Experimental Setups

Normalized dose. The pencil beam, emerging from the vacuum exit window and traversing the medical beamline detectors (*22*), was monitored by a 21×21 cm^2 parallel plate ionization chamber operated in air (further referred to as the reference chamber) (*23*) before impinging on the target. A second parallel plate ionization chamber (further referred to as the measurement chamber), equipped with a 20-mm polymethylmethacrylate (PMMA) slab placed in front to mitigate material-dependent electron build-up (*24*), was used to measure the energy loss after the target (Fig. 1A). The chamber readout was performed via Keithley® Model 6517 Electrometers (Tektronix Inc., Beaverton, OR) controlled by a LabVIEW application (DMC, Chicago, IL) and synchronized to the extraction cycle (spill) of the synchrotron.

Attenuation of the primary beam flux. A 1-mm BC400-like plastic scintillator (further referred to as START) was used to measure the flux and energy loss of all particles impinging on the target and to trigger the data acquisition. For valid scintillator signals, all emerging fragments as well as the surviving primary particles after the target were measured by a dE/E telescope consisting of a hexagonal 5-mm BC400-like plastic scintillator with an inscribed radius of 5 cm and a 23-cm-long hexagonal BaF_2 scintillator crystal with an inscribed radius of 3 cm. The detector readout was done via QDCs using a VME data acquisition system, as described by Rovituso *et al*. (*25*). A scheme of the experimental setup is shown in Fig. 1B.

Neutron ambient dose equivalent. A 21×21 cm^2 air-filled parallel plate ionization chamber was placed 80 cm downstream of the vacuum window to monitor the proton flux (Fig. 1C). After impinging on the targets (placed 75 cm further downstream), the produced neutrons were measured by three moderator-type neutron ambient dosimeters (Bonner spheres) with an additional lead layer for energy range extension (*26*). The dosimeters were equipped with Harshaw thermoluminescent dosimetry (TLD) cartridges ($2 \times$ TLD 600 + $2 \times$ TLD 700 per cartridge) at 15, 65, 90 degrees with respect to the primary beam direction and at a distance of 100 cm (measured from target center to the detector center). All TLDs were individually calibrated beforehand with a ^{137}Cs source. The conversion factor from photon absorbed dose to neutron ambient dose equivalent was obtained by measurements with a ^{241}AmBe neutron source.

To keep the induced radioactivity in the wall at a moderate level, the protons transmitted through the targets were stopped within a 35-cm thick water beam dump located at a distance of 380 cm from the exit window. The ambient neutron dose contributions originating from the water beam dump were assessed in a separate no-target run.

Targets

All experiments were performed using the same set of targets. Different polyethylene thicknesses were obtained by shifting two polyethylene wedges of 24.6-cm length, 10-cm width and a varying thickness between 0.5 mm and 7.02 cm each. The adjusted total thicknesses were measured using a caliper. Different combinations of 200.5 mm $\times$ 200.5 mm $\times$ 2.1 mm Al type 2024 plates were used to obtain the desired aluminum thicknesses.

Downloaded From: https://bioone.org/journals/Radiation-Research on 02 Nov 2024
Terms of Use: https://bioone.org/terms-of-use Access provided by University College London

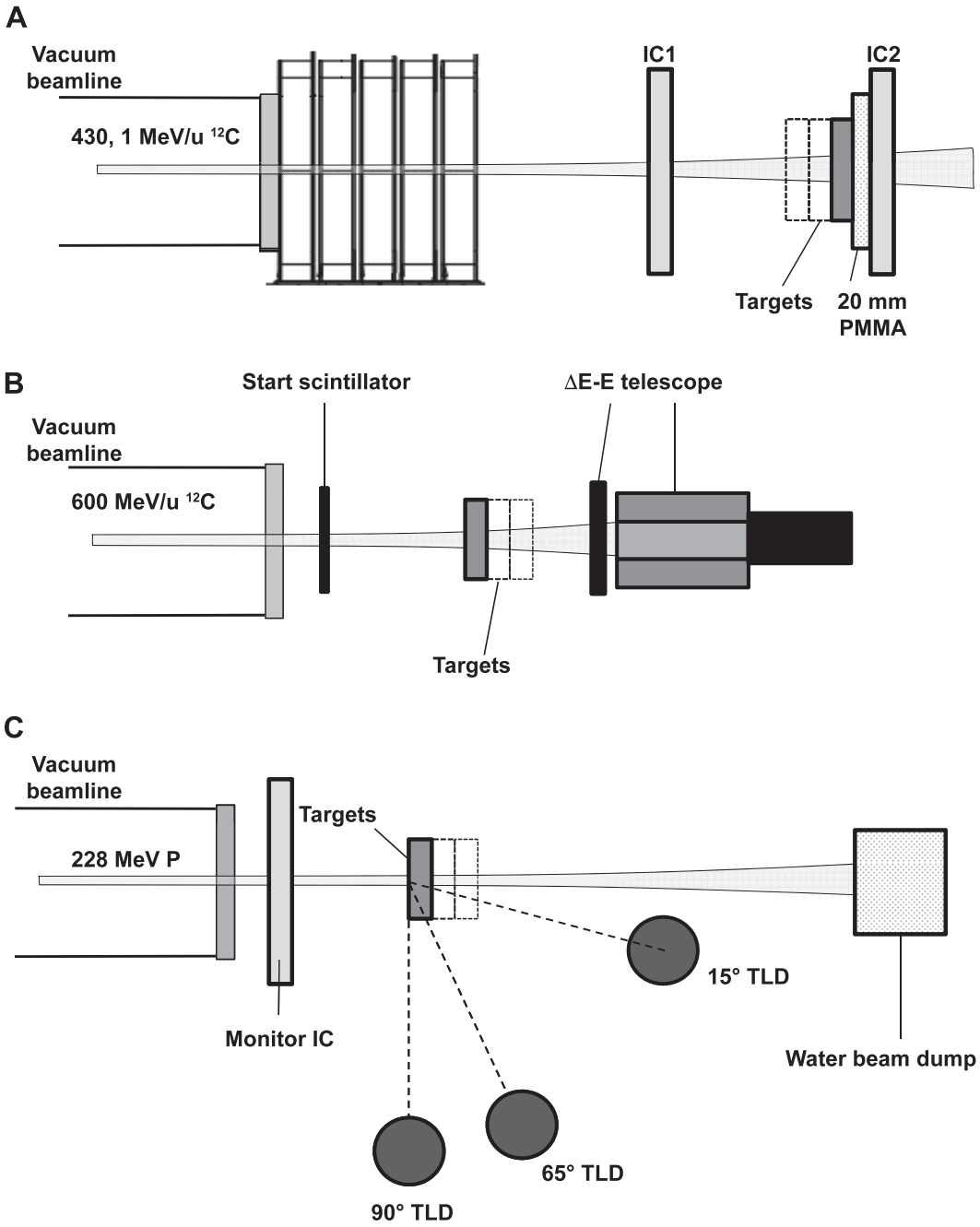


FIG. 1. Schematics of all experimental setups used in this work. Panel A: Normalized dose setup. Panel B: Attenuation of the primary beam flux. Panel C: Neutron ambient dose measurement.

LiH targets were prepared by cold-pressing LiH powder (CAS no. 7580-67-8, 92.5% ^{7}Li, 7.5% ^{6}Li; Alfa Aesar, Haverhill, MA) in a steel mold of 65-mm diameter with a mechanical pressure of 13 tons. The starting material was analyzed by X-ray diffraction in transmission geometry [PANalytical X'Pert (*27*) (Malvern Panalytical, Malvern, UK) with Cu Kα radiation] using a quartz capillary as a sample holder. Only diffraction peaks due to the LiH cubic phase were observed. The resulting weight and size of each pellet was determined by a precise scale and a caliper. The corresponding area density of the 10 investigated LiH pellets varied between 0.581 and 0.608 g/cm^2. The error on the area density was conservatively estimated to be 4% due to the inhomogeneous thicknesses at different measurement positions caused by the uneven surface of each pellet.

The pellets were then sealed inside polyethylene bags to prevent contact with moisture. All procedures were performed inside a glove box, with monitored oxygen exposure. The area density of the protective polyethylene bags (front and back foil together) was measured to be 18 mg/cm^2. The protective polyethylene bag of each pellet had an area density of approximately 3% of the LiH density and thus contributed to the measurement results. Therefore, it was considered as a separated target layer in the Monte Carlo simulations.

Downloaded From: https://bioone.org/journals/Radiation-Research on 02 Nov 2024
Terms of Use: https://bioone.org/terms-of-use Access provided by University College London

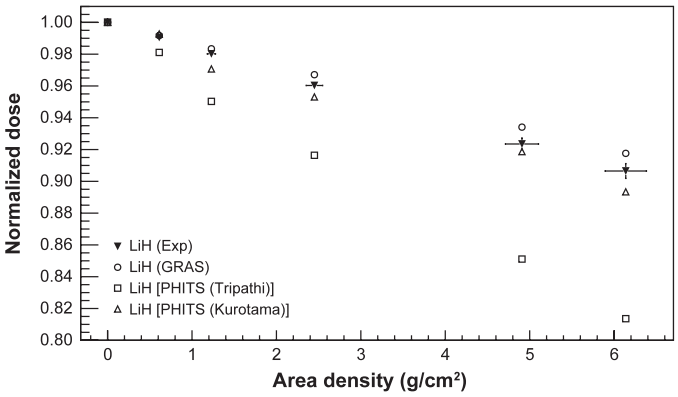


FIG. 2. Measured normalized dose values for 430 MeV/u ^{12}C traversing LiH targets of different thicknesses as well as corresponding simulations. Experimental error bars are partially hidden by the symbols.

Here, the LiH area density values plotted in Figs. 1–6 include the sum of the contribution from the LiH pellets and from the polyethylene bags.

Monte Carlo Simulations

All measured data were compared to predictions from the Monte Carlo codes PHITS (*7, 8*) and Geant4 Radiation Analysis for Space (GRAS) (*28*).

PHITS is a particle transport simulation code, developed by the Japan Atomic Energy Agency (JAEA) and several institutes in Japan and Europe, and is widely used in various research fields such as medical physics, accelerator design and space radiation physics (*29, 30*). The data presented here were obtained using PHITS version 2.82 with JQMD-2.0 (*31*) and the Statistical Multi-fragmentation Model (SMM) (*32*). In addition to the standard Tripathi-based total reaction cross-section model (*33*), further referred to as PHITS (Tripathi), all simulations were also done using the hybrid Kurotama model (*34*), further referred to as PHITS (Kurotama).

GRAS is a modular toolkit developed by the European Space Agency (ESA) for space radiation simulations and is based on the Monte Carlo code Geant4 (*9–11*). The data presented here were obtained using GRAS version 3.2 with Geant4 version 10.01 and the QBBC physics list with electromagnetic (EM) option 3, which showed

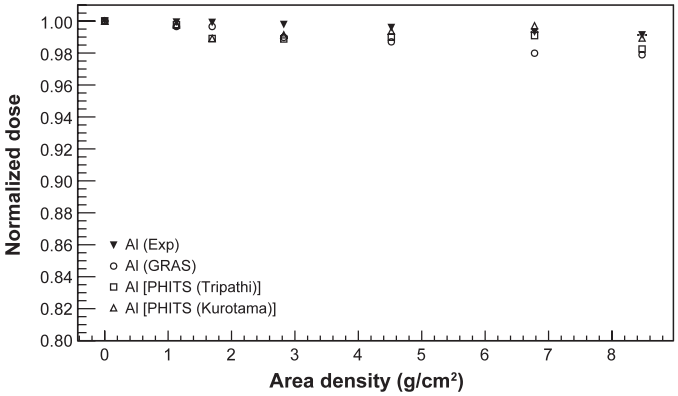


FIG. 3. Measured normalized dose values for 430 MeV/u ^{12}C traversing Al targets of different thicknesses as well as corresponding simulations. Experimental error bars are partially hidden by the symbols.

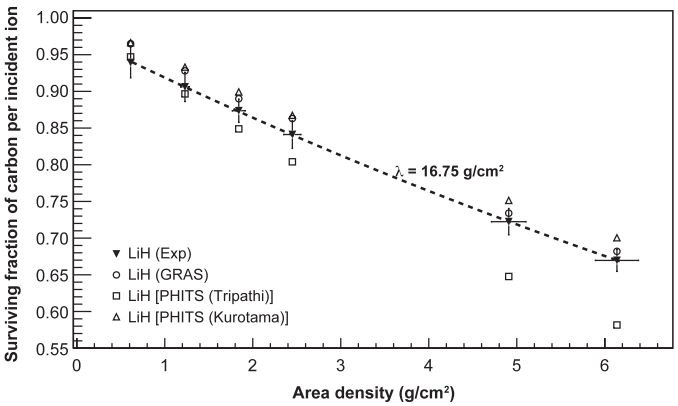


FIG. 4. Attenuation of the primary beam (charge-changing) and corresponding simulations for 600 MeV/u ^{12}C after LiH targets of different thicknesses. Experimental values are fitted with an exponential regression line (dashed). Experimental error bars are partially hidden by the symbols.

a good agreement to experimental results in the energy range between 100 MeV/u and 1.5 GeV/u (*35*).

Both Monte Carlo codes ran similar geometries and scoring methods. Detectors, targets and air gaps were fully modeled. The complex medical nozzle at HIT was substituted by a 1.8-mm water-equivalent layer (*22*). For the neutron ambient dose simulation, the geometry of the experimental room was also reproduced in full detail. To allow for a direct comparison of the simulation results with the experimental data, the response function of the neutron ambient dosimeter (*36*) was applied to the neutron spectra scored at the measurement positions.

The total number of simulated events for both Monte Carlo codes was chosen to keep the relative statistical error always below 1%.

Data Analysis

A detailed description of the data analysis performed for each type of measurement is given below.

Normalized dose. The ratio of the charge collected by the measurement- and reference chambers was calculated for each target and scaled to the same ratio obtained in a dedicated no-target run using only the 20-mm PMMA buildup plate. The normalized dose D_{norm} is then defined as

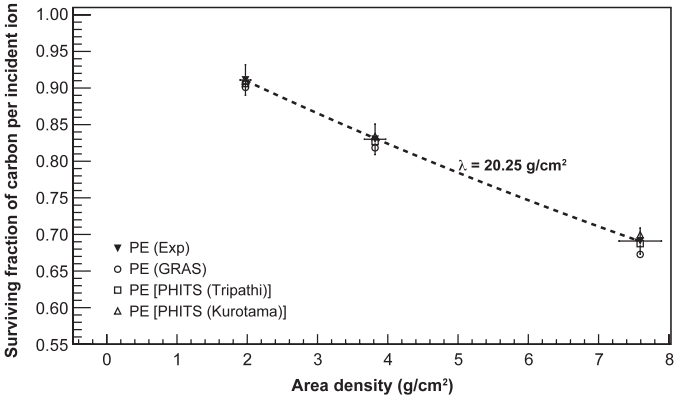


FIG. 5. Attenuation of the primary beam (charge-changing) and corresponding simulations for 600 MeV/u ^{12}C after PE targets of different thicknesses. Experimental values are fitted with an exponential regression line (dashed). Experimental error bars are partially hidden by the symbols.

Downloaded From: https://bioone.org/journals/Radiation-Research on 02 Nov 2024
Terms of Use: https://bioone.org/terms-of-use Access provided by University College London

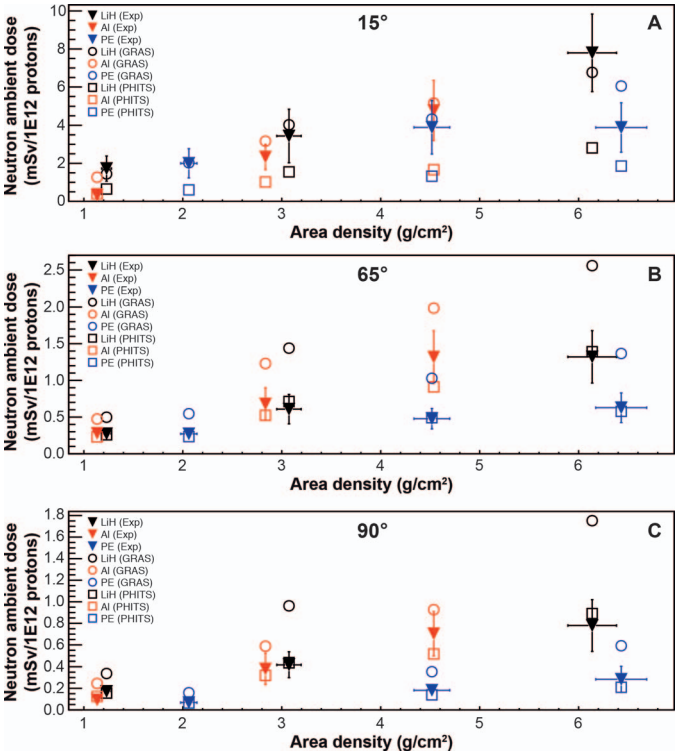


FIG. 6. Neutron ambient dose measured at 15° (panel A), at 65° (panel B) and at 90° (panel C) for PE, LiH and Al. Experimental error bars are partially hidden by the symbols.

$$D_{norm} = \frac{\left(\frac{D_{OUT}}{D_{IN}}\right)_t}{\left(\frac{D_{OUT}}{D_{IN}}\right)_0},$$

where the indices t and 0 indicate the target measurements and the no-target measurements, respectively.

The dark current of both ion chambers was monitored as an indication of the system stability and resolution and was typically found to be below 1‰ of the beam signal. However, as the measurement time was the same for all runs, the data did not require a correction to consider the dark current contribution. Due to the relatively high number of particles per spill and the overall charge integral, statistical fluctuations were negligible. Each data point was measured multiple times ($\geq$4) and averaged. Uncertainties induced by chamber effects (e.g., recombination) were conservatively estimated to 0.5%.

Attenuation of the primary beam flux. Events, for which the START signal recorded by the QDC was within three standard deviations from the primary beam peak, were classified as valid and included in the offline analysis. The identification of all carbon isotopes detected after the target was performed through a handmade graphical selection in the 2D-dE-E-plots. In this configuration, the experimental setup could not provide accurate results for low-Z ($Z < Z_{primary}$) fragments due to the typically higher multiplicity of these events. To understand if valid primary particles were not detected due to the limited size of the BaF_2 crystal or other geometrical reasons, the amount of pedestal events in the BaF_2 was correlated to valid primary beam signals in the VETO. The resulting difference was found to be <1% and was then considered in the error estimation. A detailed description of the offline analysis and error estimation approach can be found in (*25*).

Neutron ambient dose equivalent. All TLDs were read out within three days postirradiation using an automated TLD reader (Harshaw 6600 Plus Automated TLD reader). The background dose was estimated from four nonirradiated cartridges and subtracted from all values. The dose contribution, originating from the water beam dump, was measured in a no-target run and appropriately subtracted from the target measurements. The neutron ambient doses were then calculated by subtracting the average of the two independent values measured by the TLD-600 from those collected by the TLD-700. The standard deviation of each averaged dose reading was used in the error estimation. Additionally, uncertainties related to the measurement of the primary proton flux and related to the TLD calibration process were considered.

RESULTS

Normalized Dose

Figures 2 and 3 show the measured and simulated normalized dose deposited by 430 MeV/u ^{12}C in different thicknesses of LiH and aluminum targets. As expected, LiH causes a stronger dose decrease per unit thickness compared to aluminum. The latter target essentially shows no significant reduction compared to unshielded particles because the dose reduction caused by nuclear processes is almost compensated by the dose increase due to electronic energy loss (*17*).

Both Monte Carlo codes can reproduce well the experimental data for the aluminum material. The normalized dose in LiH predicted by GRAS and PHITS (Kurotama) show a fairly good agreement with the data, although the former underestimates the data for thick target, while the latter overestimates it. PHITS (Tripathi), on the other hand, shows a significant overestimation of the normalized dose independent of the target thickness. To ensure that the geometry and detector responses are not biasing the Monte Carlo predictions, PHITS (Tripathi/Kurotama) and GRAS were additionally benchmarked against Bragg curve measured in PMMA for the same primary beam energy and showed a reasonable agreement.

Attenuation of the Primary Beam Flux

Experimental and simulated results of the primary beam attenuation of 600 MeV/u ^{12}C ions in LiH and PE targets of various thicknesses are shown in Figs. 4 and 5, respectively. As expected, the experimental values show a greater loss of carbon ions in LiH than in PE at similar area densities, indicating a stronger nuclear fragmentation power of the former. Fitting the data with exponential regression lines yields an estimate of the attenuation length, equal to the slope of the exponential function. The values found are $\lambda_{LiH} = 16.75 \pm 2.25$ g/cm^2 and $\lambda_{PE} = 20.25 \pm 2.25$ g/cm^2.

Polyethylene is a widely used material in particle therapy and thus, high-quality data are available in the literature, e.g., for treatment planning system validation. For this reason all simulations reproduce the PE measurements well. The results obtained with LiH, on the other hand, are only reproduced well by GRAS and PHITS (Kurotama), while

Downloaded From: https://bioone.org/journals/Radiation-Research on 02 Nov 2024
Terms of Use: https://bioone.org/terms-of-use Access provided by University College London

PHITS (Tripathi) heavily overestimates the loss of primary carbon for increasing target thickness, resulting in a steeper gradient in the attenuation curve. As beam spot size, beam envelope and detector geometry represent possible sources for the loss of e.g., strongly scattered primary particles, a robustness analysis was performed with all used codes by slightly varying all the parameters mentioned above in the Monte Carlo simulation. These tests showed neither unexpected effects nor critical sensitivities to geometric parameters and thus gave no hint of any underlying problems in the simulations itself.

Neutron Ambient Dose

Figure 6 shows the neutron ambient dose measured at three different angles for various thicknesses of LiH, Al and PE bombarded by 228-MeV protons. The corresponding Monte Carlo simulations are plotted too. All data have been normalized to the number of primary protons impinging on the target. Both total reaction cross-section models of PHITS (Tripathi and Kurotama) predicted very similar results, and therefore, only the PHITS (Kurotama) values are indicated. The neutron ambient dose increases with increasing target thickness and decreases with increasing angle. The measurement shows that PE yields the lowest neutron ambient dose per target thickness and angle. However, the PE target geometry might cause self-shielding effects, and therefore, an additional neutron ambient dose reduction at larger angles could be observed. At 15 degrees, aluminum shows a lower neutron ambient dose compared to LiH at a similar area density. At larger angles, the neutron ambient dose caused by aluminum and LiH is comparable and typically around a factor 2 higher than polyethylene.

PHITS reproduces the experimental data well at 65 and 90 degrees for all tested materials and thicknesses but underestimates the measurements for small angles by a factor of 2.5. GRAS, however, predicts well the behavior of most materials and thicknesses at 15 degrees but overestimates the neutron ambient dose at larger angles by a factor of 2.

DISCUSSION

The findings presented in this article for neutron ambient dose can be directly used for calculations in a space environment context, whereas the choice of intermediate-energy ^{12}C ions for the normalized dose and attenuation of the primary beam measurements may not appear as relevant as a representative of the GCR environment and thus requires a dedicated discussion.

Carbon ions show a high relative abundance in the CGR environment compared to other ions in the intermediate charge region (*1*) and thus represent a reasonable representative of that part of the spectrum. Moreover, carbon ions are currently one of the available options for treating tumors. In this context, Monte Carlo codes are specifically benchmarked using ^{12}C and energies of some hundred MeV/u. Therefore, ^{12}C as primary radiation can be considered as a best-case scenario for the used Monte Carlo codes.

Figures 2–5 show that both codes reproduce well the behavior of the reference materials (aluminum and polyethylene) in terms of normalized dose and attenuation of the primary beam. One major reason for the good Monte Carlo performance with the reference materials is the comparably high quantity and quality of cross-section data due to carbon-ion therapy [e.g., (*37*)]. Overall, GRAS predicts fairly well all measurements with LiH targets. The agreement between experimental data and PHITS strongly depends on the used total reaction cross-section model. PHITS (Tripathi) overestimates the number of charge-changing reactions in LiH, leading to a higher attenuation of the beam and thus to a higher reduction in dose due to the lower dose deposition of secondaries compared to the primary particles. PHITS (Kurotama) instead shows a reasonable agreement to the presented data. To check that the reported discrepancies between simulated and measured data did not originate from geometrical errors or from the detector response, both Monte Carlo codes were benchmarked against a measured Bragg curve of 430 MeV/u ^{12}C impinging on PMMA.

As shown in Fig. 6, GRAS generally overestimates the total neutron ambient dose while PHITS underestimates it. A similar behavior was also observed by Robert *et al*. (*38*) for Geant4/GRAS and in (*39*) for PHITS. To better understand the behavior of the Monte Carlo codes, additional simulations were performed using the built-in neutron ambient dose scoring method instead of the ambient dosimeter response function. Furthermore, simplified geometries were simulated to estimate the impact of the experimental room on the results. These simulations showed only marginal improvements on the ratio of forward-to-lateral neutron ambient dose for both codes. To verify that the Bonner sphere located at 15 degrees was not hit by scattered protons, and thus having an overresponse, additional envelope as well as scattering measurements were performed and successfully reproduced by both codes. Therefore, it is unlikely that a significant amount of scattered protons could have contributed to the values measured at 15 degrees. An agreement between measured and simulated neutron doses within a factor 2 represents the typical accuracy for Monte Carlo codes in neutron radiation fields (*36*). For standard terrestrial shielding calculations these uncertainties are compensated for by making very conservative design decisions for e.g., accelerator shielding. This approach is not feasible in a space radiation environment, where any additional load is very expensive. Thus, a better precision is needed to fully elucidate the reason for these discrepancies and to improve the neutron production models. Especially for cases where a conservative evaluation can limit the project, additional and high-quality double-differential neutron production cross-section

Downloaded From: https://bioone.org/journals/Radiation-Research on 02 Nov 2024
Terms of Use: https://bioone.org/terms-of-use Access provided by University College London

measurements using similar target materials and energies are required.

In addition to its intrinsic instability, LiH demonstrated an increased shielding performance per unit mass compared to polyethylene (the reference material) for light ions at intermediate energies. This behavior is clearly indicated by the lower attenuation length measured for 600 MeV/u ^{12}C in LiH compared to polyethylene, which is approximately 20%. Additional dose reduction measurements of LiH bombarded with light and heavy ions of approximately 1 GeV/u, which are presented elsewhere (*17*), showed similar results. Therefore, if the chemical and mechanical properties of LiH can be controlled, its usage can lead to a better protection against space radiation in manned missions and to substantial cost savings, due to its higher shielding performance even at smaller masses compared to PE.

CONCLUSIONS

Experimental data on normalized dose, attenuation of the primary beam flux and measurements of neutron ambient dose equivalent have been presented for protons and ^{12}C ion beams interacting with LiH and reference materials and have been compared to the predictions of the Monte Carlo codes PHITS and GRAS.

Both codes showed reasonable agreement in normalized dose and attenuation of the primary beam flux measured in the reference materials. PHITS (Tripathi) overestimates the fragmentation of ^{12}C ions in LiH and, in effect, overestimates its dose reduction capability. Predictions from GRAS and PHITS (Kurotama) are closer to the presented data but still show deviations, especially at increasing target thicknesses. The neutron ambient dose results are overestimated by GRAS at larger angles, whereas PHITS underestimates the measurements, especially in the forward directions (smaller angles). The experiments presented in this work indicate that for light systems the standard Tripathi-based total reaction cross section model of PHITS should be used with care.

The experimental data presented here prove that LiH is a promising shielding material candidate for heavy ions, due to its high nuclear fragmentation power even at the cost of an increased secondary neutron production. However, further experiments performed at higher energies and with heavier ions are necessary to fully assess the shielding capabilities of LiH and to benchmark the Monte Carlo codes, especially for nonstandard materials. The target quality should also be improved to decrease the significant contribution of the area density to the overall experimental error.

ACKNOWLEDGMENTS

This work was supported by the European Space Agency (ESA) under contract no. TEC-EES/2014.68/AM (ROSSINI-2). We are grateful to the Radiation Protection Division of GSI for providing the neutron measurement equipment and help with the TLDs, as well as the accelerator crews of GSI, HIT and PTC.

Received: April 23, 2018; accepted: October 31, 2018; published online: November 30, 2018

REFERENCES

1. Simpson JA. Elemental and isotopic composition of the galactic cosmic rays. Ann Rev Nucl Part Sci 1983; 33:323–82.
2. Shea MA, Smart DF. A summary of major solar proton events. Sol Phys 1990; 127:297–320.
3. Chancellor JC, Scott GBI, Sutton JP. Space Radiation: The number one risk to astronaut health beyond low earth orbit. Life 2014; 4:491–510.
4. Durante M, Cucinotta F. Physical basis of radiation protection in space travel. Rev Mod Phys 2011; 83:1245–81.
5. Zeitlin C, La Tessa C. The role of nuclear fragmentation in particle therapy and space radiation protection. Front Oncol 2016; 6:65.
6. Mansur LK, Frame BJ, Callego SB, Guetersloh SB, Johnson JO, Klett JW, et al. Assessment of shielding material performance for deep space missions. Mater Res Soc Symp Proc 2005; 851:1–6.
7. Niita K, Sato T, Iwase H, Nose H, Nakashima H, Sihver L. PHITS—a particle and heavy ion transport code system. Radiat Meas 2006; 41:1080–90.
8. Iwamoto Y, Sato T, Hashimoto S, Ogawa T, Furuta T, Abe S, et al. Benchmark study of the recent version of the PHITS code. J Nucl Sci Technol 2017; 54:1–19.
9. Agostinelli S, Allison J, Amako K, Apostolakis J, Araujo H, Arce P, et al. Geant4–a simulation toolkit. Nucl Instr Meth Phys Res A 2003; 506:250–303.
10. Allison J, Amako K, Apostolakis J, Araujo H, Dubois PA, Asai M, et al. Geant4 developments and applications. IEEE Trans Nucl Sci 2006; 53:270–8.
11. Allison J, Amako K, Apostolakis J, Arce P, Asai M, Aso T, et al. Recent developments in Geant4. Nucl Instr Meth Phys Res A 2016; 835:186–225.
12. Schimmerling W, Wilson JW, Cucinotta F, Kim MY. Requirements for simulating space radiation with particle accelerators. (https://go.nasa.gov/2AOKTD9)
13. Durante M. Space radiation protection: Destination Mars. Life Sci Sp Res 2014; 1:2–9.
14. Atwell W, Rojdev K, Liang D, Hill M. Metal hydrides, MOFs, and carbon composites as space radiation shielding mitigators. Int Conf Environ Syst 2014; Paper No. 103. (https://go.nasa.gov/2JJtE9o)
15. Rojdev K, Atwell W. Hydrogen and methane loaded shielding materials for mitigation of galactic cosmic rays and solar particle events. Gravitational Sp Res 2015; 3:59–81.
16. Rojdev K, Atwell W. Investigation of lithium metal hydride materials for mitigation of deep space radiation. Int Conf Environ Syst 2016. (https://go.nasa.gov/2QnG5Ki)
17. Giraudo M, Schuy C, Weber U, Rovituso M, Giovanni S, Norbury JW, et al. Accelerator-based tests of shielding effectiveness of different materials and multilayers fusing high energy light and heavy ions. Radiat Res 2018; 191.
18. Guetersloh S, Zeitlin C, Heilbronn L, Miller J, Komiyama T, Fukumura A, et al. Polyethylene as a radiation shielding standard in simulated cosmic-ray environments. Nucl Instr Meth Phys Res B 2006; 252:319–32.
19. Welch FH. Lithium hydride: a space age shielding material. Nucl Eng Des 1974; 26:444–60.
20. Poeth, D. The evaluation of lithium hydride for use in a space nuclear reactor shield, including a historical perspective; 2018. (https://inis.iaea.org/search/search.aspx?orig_q=RN:37089053)
21. Thangavelautham J, Strawser D, Cheung MY, Dubowsky S. Lithium hydride powered PEM fuel cells for long-duration small

Downloaded From: https://bioone.org/journals/Radiation-Research on 02 Nov 2024
Terms of Use: https://bioone.org/terms-of-use Access provided by University College London

mobile robotic missions. Proc - IEEE Int Conf Robot Autom 2012; 415–22.

22. Parodi K, Mairani A, Brons S, Hasch BG, Sommerer F, Naumann J, et al. Monte Carlo simulations to support start-up and treatment planning of scanned proton and carbon ion therapy at a synchrotron-based facility. Phys Med Biol 2012; 57:3759–84.

23. Stelzer H, Voss B. Ionization chamber for ion beams and methods for monitoring the intensity of an ion beam. US Patent US 6,437,513 B1; 2002. (https://bit.ly/2APv1k4)

24. Carlsson CA, Carlsson GA. Proton dosimetry with 185 MeV protons. Health Phys 1977; 33:481–4.

25. Rovituso M, Schuy C, Weber U, Brons S, Cortes-Giraldo MA, La Tessa C, et al. Fragmentation of 120 and 200 MeV u(−14)He ions in water and PMMA targets. Phys Med Biol 2017; 62:1310–26.

26. Fehrenbacher G, Gutermuth F, Kozlova E, Radon T, Schuetz R. Neutron dose measurements with the GSI ball at high-energy accelerators. Radiat Prot Dosimetry 2007; 125:209–12.

27. Malvern Panalytical. X-ray diffractometers (XRD). (https://bit.ly/2F5N3CD)

28. Santin G, Ivanchenko V, Evans H, Nieminen P, Daly E. GRAS: a general-purpose 3-D Modular Simulation tool for space environment effects analysis. IEEE Trans Nucl Sci 2005; 52:2294–9.

29. Aghara SK, Sriprisan SI, Singleterry RC, Sato T. Shielding evaluation for solar particle events using MCNPX, PHITS and OLTARIS codes. Life Sci Sp Res 2015; 4:79–91.

30. Zeitlin C, Guetersloh S, Heilbronn L, Miller J, Fukumura A, Iwata Y, et al. Fragmentation cross sections of 290 and 400 MeV/nucleon C12 beams on elemental targets. Phys Rev C 2007; 76:014911.

31. Ogawa T, Sato T, Hashimoto S, Satoh D, Tsuda S, Niita K. Energy-dependent fragmentation cross sections of relativistic (12)C. Phys Rev C 2015; 92:1–14.

32. Ogawa T, Sato T, Hashimoto S, Niita K. Analysis of multi-fragmentation reactions induced by relativistic heavy ions using the statistical multi-fragmentation model. Nucl Instruments Methods Phys Res Sect A Accel Spectrometers, Detect Assoc Equip 2013; 723:36–46.

33. Tripathi RK, Cucinotta FA, Wilson JW. Accurate universal parameterization of absorption cross sections III - light systems. Nucl Instr Meth Phys Res B 1999; 155:349–56.

34. Sihver L, Kohama A, Iida K, Oyamatsu K, Hashimoto S, Iwase H, et al. Current status of the ''Hybrid Kurotama model'' for total reaction cross sections. Nucl Instr Meth Phys Res B 2014; 334:34–9.

35. Ivantchenko A V, Ivanchenko VN, Molina JQ, Incerti SL. Geant4 hadronic physics for space radiation environment. Int J Radiat Biol 2012; 88:171–5.

36. Horst F, Fehrenbacher G, Zink K. On the neutron radiation field and air activation around a medical electron linac. Radiat Prot Dosimetry 2017; 174:147–58.

37. Haettner E, Iwase H, Kraemer M, Kraft G, Schardt D. Experimental study of nuclear fragmentation of 200 and 400 MeV/u (12)C ions in water for applications in particle therapy. Phys Med Biol 2013; 58:8265–79.

38. Robert C, Dedes G, Battistoni G, Bohlen TT, Buvat I, Cerutti F, et al. Distributions of secondary particles in proton and carbon-ion therapy: a comparison between GATE/Geant4 and FLUKA Monte Carlo codes. Phys Med Biol 2013; 58:2879–99.

39. Iwamoto Y, Taniguchi S, Nakao N, Itoga T, Nakamura T. Measurement of neutron production spectra at the forward direction from thick graphite, aluminum, iron and lead targets bombarded by 250 MeV protons (JAEA-Conf–2006-009). In: Tahara Y, Fukahori T, editors. Japan: International Nuclear Information System; 2006. (https://bit.ly/2SQgefN)

Downloaded From: https://bioone.org/journals/Radiation-Research on 02 Nov 2024
Terms of Use: https://bioone.org/terms-of-use Access provided by University College London